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Research paper

Data-centric explainable artificial intelligence techniques for cyber-attack detection in microgrid networks

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ABSTRACT

This article presents a strategy for pre-processing data and interpreting the performance of advanced machine learning (ML) models to improve data quality and modify the models based on identified problems during the iterative development of classification models. The process consists of three main stages aimed at maximizing the performance of ML models. Stage-1 involves data collection and preprocessing. Stage-2 focuses on improving data quality and extracting features. The extracted features are further refined using hyperparameters in Stage-3. Additionally, the framework utilizes SHapley Additive exPlanations (SHAP) for explainable artificial intelligence (XAI) to elucidate the learning process of ML models and, therefore, make the model's behavior interpretable. As an example, this innovative process has been applied to cyber-attack detection in microgrid networks. In Stages-1, a CIGRE low-voltage microgrid network is simulated for data collection during cyber-attacks, followed by data pre-processing. In Stage-2, data augmentation is performed using SMOTE and ENN, and feature extraction is carried out using the Boruta Python package. In Stage-3, hyperparameters are tuned using the TPE algorithm. The results demonstrate the effectiveness of this approach, enabling the model to progressively enhance its predictive capacity through each stage.

1. Introduction

As energy systems evolve into interconnected, digitalized infrastructures, they face an escalating threat landscape characterized by sophisticated cyber-physical attacks. Smart grids, with their reliance on digital communication and automated control, are particularly vulnerable to cybersecurity risks, necessitating advanced detection and mitigation strategies. Machine learning, blockchain, and federated learning have emerged as pivotal technologies in bolstering these systems' security, enabling real-time threat detection, adaptive response mechanisms, and privacy-preserving data storage. Recent research highlights the breadth of approaches to safeguarding energy infrastructure, spanning machine learning models for anomaly detection to testbeds that assess grid resilience under simulated cyber-physical threats.

With the rapid integration of information and communication technologies, cyberspace has become indispensable in the smart grid context, particularly for operating, controlling, and monitoring physical assets related to energy production, distribution, and consumption. However, the interaction between industrial control systems and the Internet environment of IoT technology makes communication technologies in critical infrastructure applications vulnerable to cyber-attacks. A comprehensive review by Diaba et al. (2024) emphasizes the dual nature of cyber and physical vulnerabilities within modern energy systems. Their work explores existing cybersecurity measures, particularly intrusion detection mechanisms, which are foundational in detecting and responding to cyber-physical attacks. This review provides a critical assessment of these measures within the evolving landscape of renewable integration and decentralized energy storage, highlighting the need for cybersecurity strategies that can keep pace with these advancements.

Machine learning has proven indispensable in addressing the complex detection requirements of these systems. A study by Aurangzeb et al. (2024) demonstrates the application of deep learning models in detecting adversarial attacks on smart grids, with a particular focus on black-box attacks. Their approach leverages quantum voting ensemble methods to enhance detection accuracy while using blockchain technology for privacy-preserving data storage, showcasing a dual-function solution that supports both security and data integrity. Another approach, explored by Hashim et al. (2024), involves the use of stacked machine learning models to identify non-technical losses (NTLs), such as

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energy theft. This research compares the effectiveness of different model configurations, aiming to address NTLs as a pervasive issue that impacts grid efficiency and economic stability.

The extensive network traffic in Cyber-Physical Systems (CPSs) renders traditional machine learning (ML) methods for network anomaly detection less effective (Ashrafuzzaman et al., 2020). Given the significance of smart grids as vital national infrastructure, safeguarding the grid from cyber threats is crucial. In this regard, ML methods have gained prominence for detecting threats by identifying natural patterns and anomalies in data, thereby enhancing predictive accuracy (Abdullahi et al., 2022; Chen et al., 2021). Consequently, recent ML advancements, particularly those emphasizing deep learning, have successfully detected and classified anomalies at both the network and host levels.

As technology progresses, network intrusion detection methods have evolved from basic port inspection to sophisticated machine learning techniques. Traditional port-based approaches have become obsolete due to modern applications' dynamic port allocation and the rise of encrypted traffic, which undermines payload-based methods. As a result, cybersecurity experts are increasingly relying on ML and network flow features to enhance detection methods. Recent advancements in ML techniques for network anomaly detection have been well-received (Ali et al., 2023; Bitirgen and Filik, 2023). In cloud environments, the diversity of virtualized applications poses challenges that ML can effectively address by resolving network anomalies. This paper aims to provide an ML-based anomaly detector and optimal solutions to enhance detection tools' resilience.

1.1. Literature survey

Several well-established models are employed for ML-based classification problems, particularly for detecting cyber-attacks. The Bag-of-SFA-Symbols in Vector Space (BOSSVS) model, which uses the discrete Fourier transform on a time series, performs well with noisy data, though its application in cyber-attack detection is underexplored (Schäfer, 2015). Support Vector Machines (SVM), (Yan et al., 2016) modeled as a binary classification problem, can classify 'attack' or 'normal' scenarios based on available data but cannot identify the type of attack (Ahmad et al., 2021; Tuan et al., 2020). A multi-class classification SVM model using a reduced dataset (UNSW-NB15) demonstrates high false alarm rates with accuracy scores of 95.67 % and 84.32 %, respectively (Moustafa and Slay, 2015).

Authors in (Kumar et al., 2021) have focused on feature engineering to extract relevant features for cyber-attack detection. The Naïve Bayes algorithm (Venkatesh and Ranjitha, 2018) operates under a strong feature independence assumption and utilizes feature extraction and multiple validation techniques to compare model accuracy. The Ridge classifier, based on ridge regression, converts labelled data into a specific range and solves it as a regression problem (Saheed and Arowolo, 2021). Additionally, particle swarm optimization is employed to select an optimal number of relevant features.

Deep learning, a specialized subcategory of ML techniques that mimics neural networks inspired by the brain, has been applied to identify attacks. This research considers an adaptive boosting classifier (AdaBoost), which enhances classification accuracy by combining weak classifiers into a single robust classifier.

In addition to pure machine learning models, hybrid approaches are increasingly being adopted to assess system resilience against cyber threats. For instance, Alasali et al. (2024) develop a modern adaptive testbed that combines physical and co-simulation components to rigorously evaluate smart grid resilience. This testbed enables researchers to simulate cyber-physical threats and adapt protection schemes in real time, bridging the gap between theoretical detection models and practical, real-world applications.

Authors in Subba et al. (2016) have demonstrated that Artificial Neural Networks (ANN) for multi-class classification models achieve higher accuracy (around 95 %) than Naïve Bayes models (around 91.5 %). Multi-layer ANN models (Vinayakumar et al., 2019) used for various cyber-attack classifications show varying accuracy on datasets, with 5-layer ANN models achieving 78.5 % compared to 76.3 % from classical decision tree models (NSL-KDD). A GRU model (Rehman et al., 2021) developed for web-based cyber-attack detection achieves 95.11 % accuracy, slightly lower than the RNN model's 95.6 %. GRU models used for network intrusion detection attain 89 % accuracy on the NSL-KDD dataset, compared to 75.9 % with deep ANN algorithms (Tang et al., 2018). Authors in Kaur et al. (2023) presented a method for extracting features from a multidimensional feature space using convolutional and pooling neural layers. They also utilized Bayesian probabilistic methods to measure uncertainties in model parameters and achieved an accuracy of 70.96 % for GRU, 88.38 % for LSTM, and 84.86 % for CNN-Bayesian in a multi-class classification approach.

A 5-layer convolutional neural network (CNN) with two max-pooling layers and four fully connected output layers is proposed in Al-Turaiki and Altwaijry (2021) for network intrusion detection, with experiments conducted on binary and multi-class classification problems. Researchers in Jo et al. (2020) have developed a CNN-based algorithm focusing on data pre-processing techniques that convert single packet data into 28×28 images for cyber-attack detection. A multi-CNN fusion model achieves 81.33 % accuracy on the KDD test dataset. Other studies in Kim et al. (2016) use LSTM-RNN models on the KDD dataset, achieving 96.93 % accuracy. An LSTM model compared against an ANN model (Khan et al., 2021) on a consumer load dataset in Michigan shows the LSTM model's accuracy at 96.57 %, compared to the ANN model's 88.86 %.

To address the smart meter cyber-attack problem, researchers in Yao et al. (2021) use a CNN-LSTM model that fuses CNN and LSTM layers. The CNN identifies regional components, while LSTM extracts periodic feature patterns. A 3-layer CNN-LSTM model achieves 98.7 % accuracy on the KDD Cup 99 dataset (Vinayakumar et al., 2017), the highest among all other CNN-RNN and GRU combinations. Additionally, a combined CNN-LSTM algorithm is proposed in Abdallah et al. (2021) for anomaly detection in Software-Defined Networking.

The time-series transformer (TST) is a recently developed algorithm suitable for multi-class classification problems (Zerveas et al., 2021), though its application in cyber-attack detection remains unstudied.

Comprehensive reviews on cyber-attacks compiled in Ghiasi et al. (2023), Tatipatri and Arun (2024) and analyzed contemporary methods for identifying diverse cyber-attacks in smart grid systems. These range from sensor and actuator attacks to Denial of Service (DoS), False Data Injection Attacks (FDIA), Data Integrity Attacks (DIA), Resonance Attacks (ResA), Zero Dynamics Attacks (ZDA), man-in-the-middle attacks (MiTMA), and Time Delay Switch Attacks (TDSA).

Existing research (Kumar et al., 2021; Esmalifalak et al., 2017; Trivedi et al., 2022a) often focuses on binary classification problems, limiting the user's ability to identify the type of attack. Addressing this gap requires a multi-class classification problem. Class imbalance, where samples in the number of classes are unequal, leads to uncertainty and low model accuracy.

1.2. Key contribution

This research focuses on a data-centric approach to cyber-attack detection, including data collection, pre-processing, training advanced ML and deep learning models, interpretation, and data quality improvement for multi-class classification problems with hyper-parameter tuning. The key contributions are as follows:

- The study formulates a problem using a unique encoder-based transformer network to compare accuracy with the most popular ML models.
- A detailed analysis of cyber-attack detection has been extended using deep learning techniques, emphasizing model performance

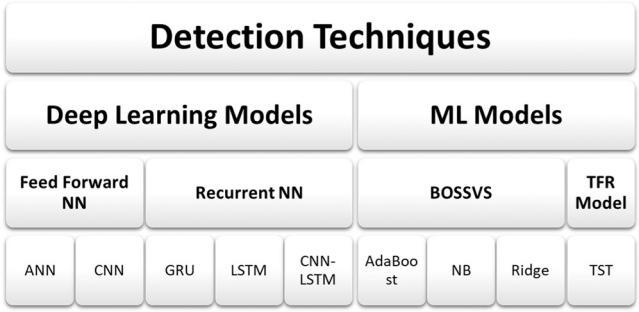


Fig. 1. Implemented Machine and Deep Learning models for Cyber-attack detection.

understanding, referred to as Explainable Artificial Intelligence (XAI). The detection techniques are presented in Fig. 1. A comprehensive comparative analysis of the different models used is presented in the subsequent section.

- A unique attribute of the proposed framework is understanding the individual model learning performance, called XAI (Barredo Arrieta et al., 2020), which helps to improve the system's cyber-attack detection capability.

Understanding model behaviour for practical applications is crucial. Complex models developed for high accuracy in ML applications pose challenges in performance interpretation. A unified framework for prediction interpretations, SHAP (Lundberg and Lee, 2017) assigns important values to feature-specific predictions and quantifies each feature's contribution to the output while considering feature interactions. Various metrics and techniques for model interpretation are discussed later in this paper.

The remainder of the article is structured as follows: Section 2 presents the structure of the CIGRE LV MG network. Section 3 introduces the data-centric classification framework. Section 4 describes the developed classification models. Section 5 discusses model interpretation. Section VI explains the model evaluation procedure. Section 6 deals with experiment results and analytical analysis. Finally, Section 7 provides the conclusion.

2. System modeling for data acquisition

Modern power electronic devices largely enable the smart grid to integrate and control diverse renewable energy sources (RES) with high efficiency and reliability. These devices play a crucial role in mitigating the challenges posed by high renewable energy adoption. Most distributed energy resources (DERs), such as photovoltaic panels, wind turbines, and battery storage systems, connect to the smart grid via inverters. Grid-tied inverters serve as the interface between DERs and the grid, equipped with advanced features to provide ancillary services.

However, the proliferation of information and communication technologies (ICTs) has increased the vulnerability of smart inverters to cyber-attacks. To comply with IEEE standard 1547–2018 (IEEE, 2018), DER devices must support at least one of IEEE 2030.5, IEEE 1815 (DNP3), or SunSpec Modbus protocols. SunSpec modbus facilitates communication within DER systems, while IEEE Std. 2030.5 is widely adopted by investor-owned utilities for monitoring and regulating smart inverters (SunSpec IEEE, 2030.5 / CSIP Certification, 2023). Utility SCADA systems commonly use IEEE 1815 to interconnect DER devices, including smart inverters, and the IEC 61850 standard, which defines information models and functionalities for DER devices, is also a popular protocol.

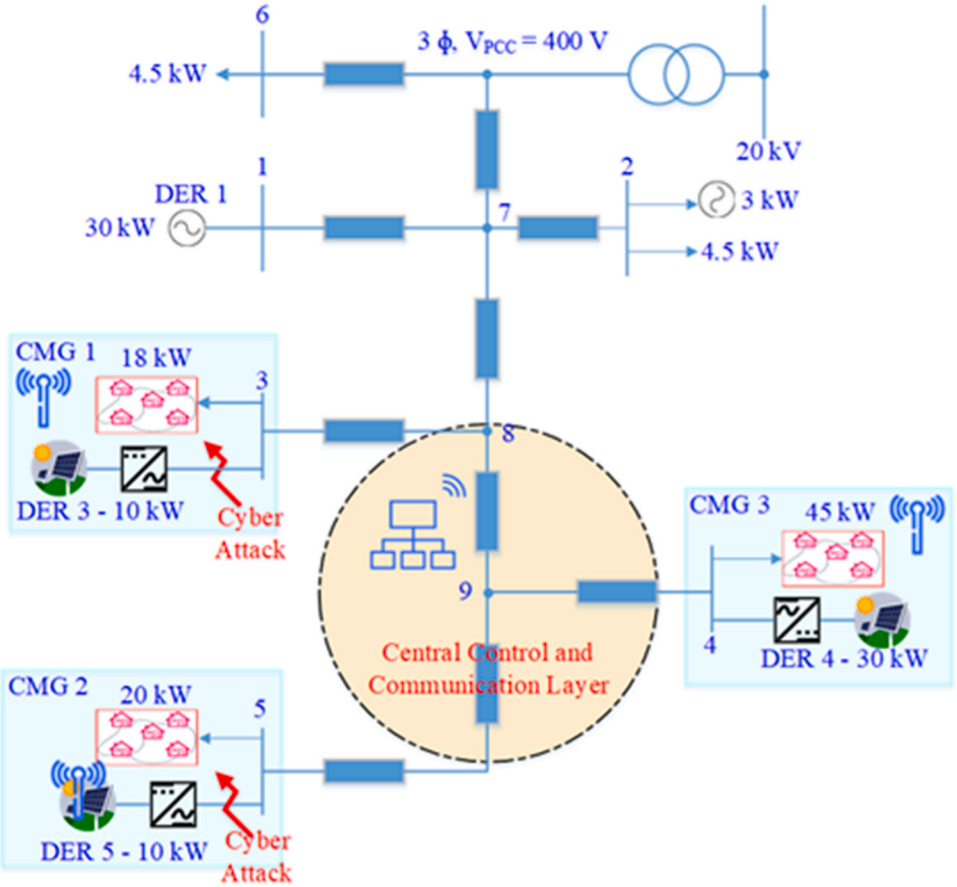


Fig. 2. Simplified CIGRE Low Voltage (LV) microgrid reference network.

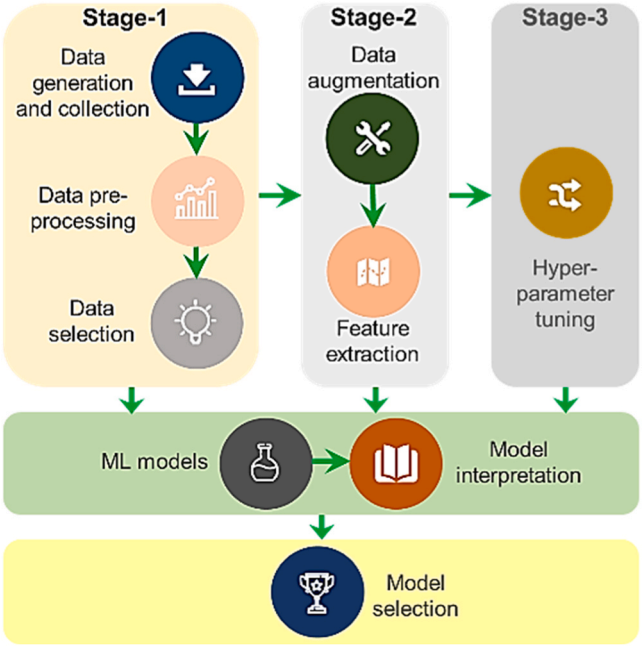


Fig. 3. Classification Framework.

Table 1
Labels of the Classifications.

Event type	label
Normal	1
Load Change (Incremental)	2
Line fault (phase 'A')	3
Load Change (Decremental)	4
Sensor Fault	5
Cyber-attack (Inverter−1)	6
Cyber-attack (Inverter−2)	7

Cyber-attacks on smart inverters can have various objectives, targeting either individual devices or the entire grid and its services. Device-level attacks aim to alter operating conditions or disrupt communication, potentially preventing grid operators from achieving objectives such as optimal power flow or minimal energy loss. Cyber-attacks can range from device-level disruptions to broader attacks affecting interactions between smart inverters and control centers or other inverters.

A comprehensive literature review reveals that cybercriminals employ multiple methods to attack smart inverters, including hardware-based attacks like firmware substitution, hall spoofing, and phase-locked loop (PLL) assaults; software-based attacks such as reconnaissance and packet replay; and communication line attack vectors like Denial of Service (DoS) and Man-in-the-Middle (MITM) attacks. These attacks threaten the fundamental, grid-support, and protective functions of inverters. Access to sensitive information, such as control parameters or critical settings, can amplify the severity of an attack. While firmware encryption complicates firmware replacement and PLL attacks require high-voltage pulse generators, hall spoofing remains the most accessible method for initiating False Data Injection Attacks (FDIAs), immediately impacting inverter functionality and protection.

The CIGRE low voltage microgrid reference network, depicted in Fig. 2, features a four-wire overhead main feeder supplying a suburban residential area. This study uses three microgrid units equipped with DERs and grid-tied inverters (GTIs) to power the system loads. GTIs operate under a droop control system, with voltage and current reference signals generated based on information from a central control and communication layer, which coordinates microgrid (MG) interactions. The outer droop-based loop stabilizes voltage and frequency, providing voltage references for the inner voltage loop. The voltage controller output serves as the current reference for the filter inductor current, with the final output being the pulse width modulation switching reference signal. To protect semiconductor switches in GTIs, a hybrid reference frame limiting (HRFL) strategy is employed, limiting inductor current reference to twice the rated current through a current limiting factor. The study system's parameters are detailed in Barredo Arrieta et al. (2020), with a nominal frequency of 50 Hz and line-to-line voltage of 0.4 kV. Simulations include events such as load changes, generation changes, line faults, sensor faults, and cyber-attacks (FDI), with the resulting data used to train models for attack-type identification.

3. Data-centric classification framework

The comprehensive three-stage classification framework, shown in Fig. 3, primarily focuses on data pre-processing (Stage-1), quality improvement (Stage-2), and hyper-parameter tuning (Stage-3), also developed by the same authors in Trivedi et al. (2022a). The performance of models across various metrics is analyzed and examined in the "Model interpretation" layer. Here is a detailed description of each stage:

3.1. Stage-1

3.1.1. Data generation and collection

At this stage, the simulation of the CIGRE reference microgrid network is carried out in a MATLAB/Simulink environment. The simulation covers a range of scenarios, including changes in load and generation, line and sensor faults, as well as cyber-attacks (FDI). Once the system characteristics are recorded, the data is analyzed using machine learning-based cyber-attack detection. Subsequently, the simulated data is stored in a mat file format on a cloud server.

3.2. Data pre-processing and data selection

After collecting the data, the corresponding mat files are converted to Python data frames to facilitate data analysis. For pre-processing, the authors have already developed a method (Trivedi et al., 2022b). Table 1 outlines the events, which are labelled as numeric digits for time series classification. Visualization and data statistics are carried out to analyze the distribution of multi-class samples and box-plot functions are used to visualize mean, median, and max value distributions.

3.3. Stage-2

3.3.1. Data augmentation

Various techniques, such as random under and over-sampling, have already been employed in many published works to address the issue of class imbalance. Random under-sampling involves sampling from the majority class until a balance is reached with the minority class, which may result in a loss of information from the majority class. Whereas, the minority class includes duplicate samples in random over-sampling cases until a balance is achieved. However, this method may lead to overfitting with respect to the minority class.

SMOTE (SunSpec IEEE, 2030.5 / CSIP Certification, 2023) addresses class imbalance by generating synthetic samples in the feature space. It uses k-nearest samples to combine over-sampling of the minority class, unlike traditional methods that operate in data space. SMOTE generates synthetic samples rather than operating in data space by performing in feature space. This approach improves classifier performance in the ROC (receiver operating characteristic) curve space compared to just under-sampling the majority class.

ENN is a method (Wilson, 1972) that removes noisy instances whose class differs from at least two of its three nearest neighbors from both majority and minority classes. The method follows a

three-nearest-neighbor rule and then uses a single-neighbor rule to classify the remaining pre-classified samples to achieve good performance.

*The **SMOTE+ENN-*** method combines SMOTE and ENN approaches simultaneously (Picek et al., 2019) First, class balance is achieved by generating synthetic samples using the SMOTE method. The ENN method then removes noisy instances from all classes, enriching the datasets with more defined class clusters. It first applies SMOTE (Synthetic Minority Over-sampling Technique) to increase minority class samples by creating new data points. For a minority class sample x_i and a randomly selected k-nearest neighbor $\hat{x}_i$, SMOTE generates a synthetic sample as:

$$x_{new} = \quad x_i + \delta(\hat{x}_i - x_i)$$

where δ is a random value between 0 and 1. If class imbalance persists, this process is repeated. Next, ENN (Edited Nearest Neighbors) removes noisy samples by checking if a sample is misclassified by its three nearest neighbors; if so, it is deleted. Finally, Mixup augments the data further by linearly combining two samples x_i and x_j from the same class as:

$\hat{x} = \lambda x_i + (1 - \lambda) x_j$where λ is sampled from a Beta distribution. This combination of SMOTE, ENN, and Mixup results in a balanced, cleaner training dataset.

Algorithm 1. Balancing Imbalance using SMOTE+ENN

Input: Imbalance Training Data
Output: Balanced Training Data
1: Apply SMOTE-ENN:
2: Randomly select x_i from minority classes
3: Identify k-nearest neighbors of x_i and randomly select one of the neighbors $\hat{x}_i$
4: Generate $x_{new} = x_i + \delta(\hat{x}_i - x_i)$ where ($\delta \in [0, 1]$)
5: **If** balancing ratio does not satisfy **then**
6: Go to step 1
7: **else**
8: Remove noise samples using ENN:
9: **for** every instance x_i **do**
10: Find the three nearest neighbors of x_j
11: **if** x_j gets misclassified by its three nearest neighbors
12: **then**
13: Delete x_j
14: **end if**
15: **end for**
16: **end if**
17: Apply Mixup:
18: Randomly select x_i and x_j from the same classes
19: Randomly sample λ from Beta distribution
20: $\hat{x} = \lambda x_i + (1 - \lambda) x_j$
21: **End**

Fig. 4 shows the plot for sample observation count, illustrating the number of samples pre-and post-processing of the SMOTE+ENN technique.

3.4. Feature extraction

The Boruta algorithm (Kursa et al., 2010) extracts essential features from a given dataset for a specific classification problem. It operates by leveraging the random forest (RF) algorithm, which is the main component of Boruta. RF ensembles weak decision trees based on the

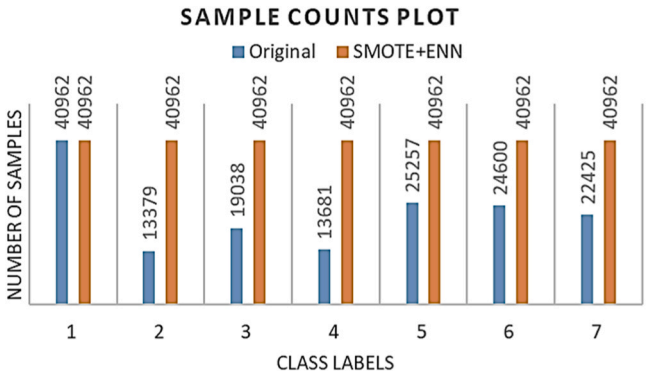


Fig. 4. Pre and post-processing of sample counts using SMOTE+ENN.

decision tree principle. One of the most important aspects of RF is the concept of majority voting. In a classification problem, each tree in the model casts a vote to classify the points, and then the point receiving the majority vote is selected as the final classification.

3.5. Stage-3

3.5.1. Hyper-parameter tuning

Hyperparameters are important characteristics of a machine-learning model. However, finding the best hyperparameters can be time-consuming in AI projects. One common method is manually adjusting the parameters and observing the model's performance through trial and error. Another method is automatic parameter search within a given search space, allowing quicker and more efficient model training with improved performance.

In this research, a tree Parzen estimator (TPE) (Bergstra et al., 2011) is used as the parameter search algorithm. The details of TPE are explained in Trivedi et al. (2022b) Nevertheless, some parameters in this paper are manually adjusted for the TST model. Using the TPE algorithm

enables fine-tuning of parameters such as learning rate, number of layers, and regularization strength. This optimization enhanced the model's adaptability to new attack patterns by allowing it to focus on features most indicative of anomalous behaviour across evolving cyber threats. By dynamically adjusting these parameters, the model achieves a balance between high predictive accuracy and resilience, ensuring consistent performance as new types of cyber-attacks emerge.

4. Classification models

The multi-class classification models for cyber-attack detection, as outlined in Fig. 1, have been implemented in this research. A brief description of the models is given below:

4.1. Machine learning (ML) models

BOSSVS (Middlehurst et al., 2019) - is a dictionary-based classifier used for time series classification. The algorithm replaces "parameter search" with randomly selected sets to improve its performance in terms of speed and accuracy. It transforms time series into histograms and then adds up all the histograms for each class to calculate a term frequencies-inverse document frequency (tf-idf) vector. When predicting a new sample class, it selects the class with the highest cosine similarity between its tf and tf-idf vectors of each class.

AdaBoost - Adaptive boosting begins by assigning equal weights to each training sample observation. It then employs multiple weak models and assigns higher weights to observations when misclassification occurs. By using several weak models and decision function boundaries achieved through multiple iterations, AdaBoost enhances the overall accuracy of the classification model.

NB classifier - Assumes that the feature values are independent of each other given the target value. This means that for a specific target instance, the probability of observing a combination of features is just the product of the probability of each feature individually (Venkatesh and Ranjitha, 2018). As a result, the classifier overlooks potential dependencies or correlations among input features, simplifying the multivariate problem into a set of univariate problems. Therefore, the NB classifier essentially overlooks potential dependencies, like correlations among input features, and simplifies the multivariate problem into a set of univariate problems.

Ridge classifier – The algorithm uses ridge regression and first transforms the target values into a specific range for multi-class problems. Then, it treats the problem as a multi-output regression task (Saheed and Arowolo, 2021). It trains the n-class classifiers using a one-versus-all approach and includes built-in cross-validation for experimentation using the Leave-One-Out Cross-Validation approach.

TST framework (Zerveas et al., 2021) - used in this research is based on the transformer encoder architecture. Here, the existing data samples are reused to train the model through unsupervised learning. The transformer models utilize a multi-headed attention mechanism to capture various correlations among input elements in the time series. Unlike other models, TST does not use a decoder, typically used for generative tasks like natural language processing, as it requires a ground truth output sequence. This makes TST more versatile for classification tasks as it only utilizes an encoder.

SVM (Yan et al., 2016) - is a powerful learning strategy-based method used for most classification problems. In this study, we are using SVM not just for binary classification but also for multi-class classification and prediction. The SVM is trained on input samples with associated target labels, which represent the classes. Input samples are processed as attribute vectors, and the SVM identifies the

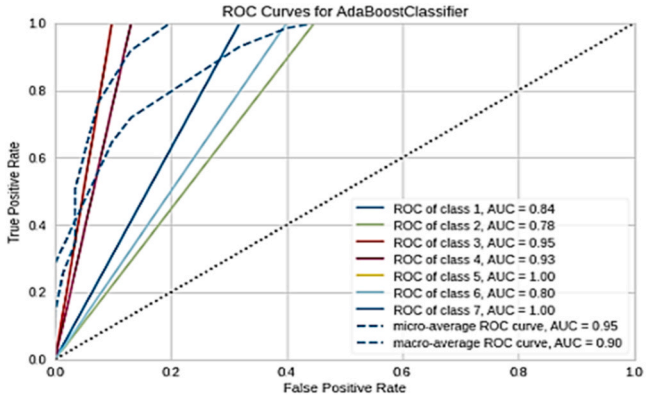


Fig. 6. ROC curve for AdaBoost classifier (Stage-1).

Table 2
Model performance at stage-1.

Models (stage 1)	Acc.	Recall	Precision	F1
Bossvs	0.77	0.77	0.86	0.75
SVM	0.88	0.86	0.92	0.88
Adaboost	0.67	0.57	0.50	0.56
NB	0.92	0.90	0.94	0.92
Ridge	0.89	0.84	0.9	0.88
TST	0.90	-	-	-
ANN	**0.68**	**0.59**	**0.6**	**0.53**
GRU	**0.89**	**0.86**	**0.85**	**0.85**
CNN-LSTM	**0.84**	**0.81**	**0.89**	**0.77**
CNN	**0.97**	**0.96**	**0.98**	**0.97**

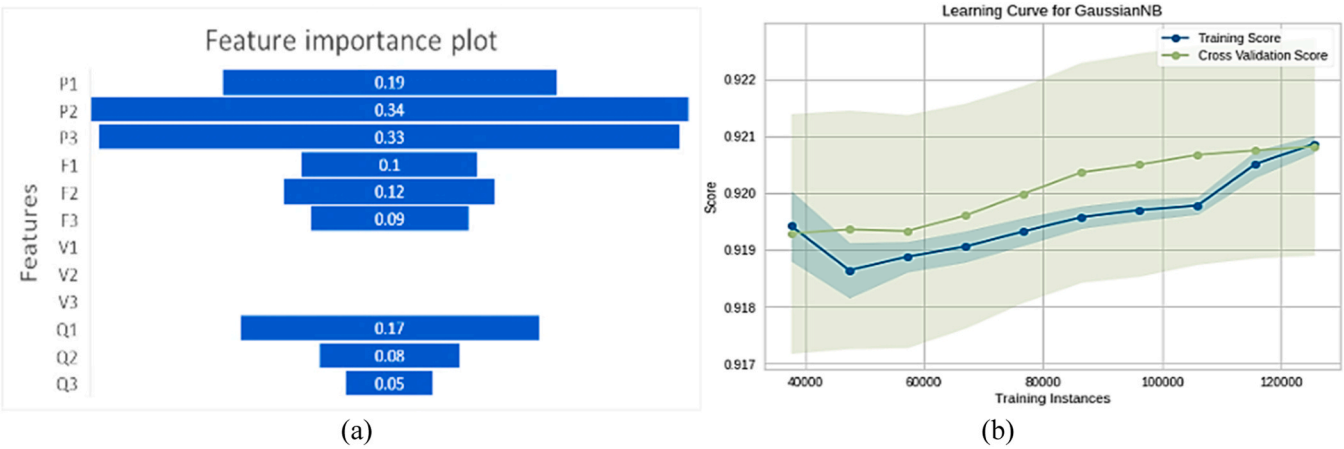


Fig. 5. (a) Score on Feature importance (max value) (b) Learning curve - based on naive Bayes algorithm (Stage-1).

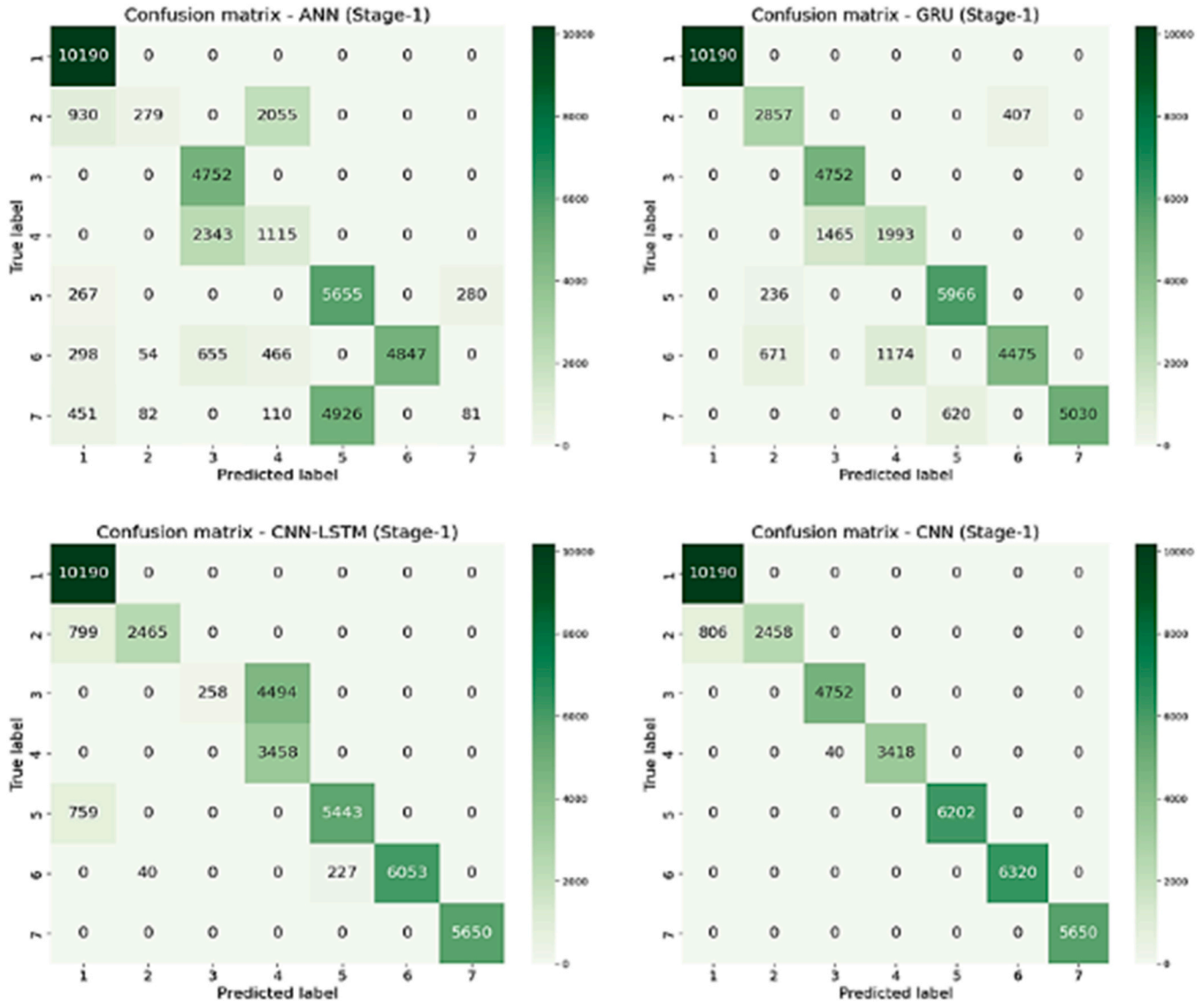


Fig. 7. Confusion matrix (Stage-1).

Table 3
Model performance (different classes' predictions).

STAGE - 1	Class Label	ANN	GRU	CNN-LSTM	CNN
Normal	**1**	✓	✓	✓	✓
Load Change (Incremental)	**2**	×	×	×	×
Line fault (phase 'A')	**3**	✓	✓	×	✓
Load Change (Decremental)	**4**	×	×	✓	✓
Sensor Fault	**5**	×	×	×	✓
Cyber-attack (Inverter−1)	**6**	×	×	×	×
Cyber-attack (Inverter−2)	**7**	×	×	✓	✓

appropriate hyperplane by separating input samples and being farthest from the convex hulls. For multi-class classification, we are using the 'original one-vs-one (ovo)' decision function shape.

4.2. Deep-learning (DL) models

ANN - In practical applications with a massive amount of data, ANN can be useful in identifying recurring meaningful arrangements in the data. ANNs gain insights about the data by modifying weights and biases by iterating over batches of samples and adjusting the parameters to

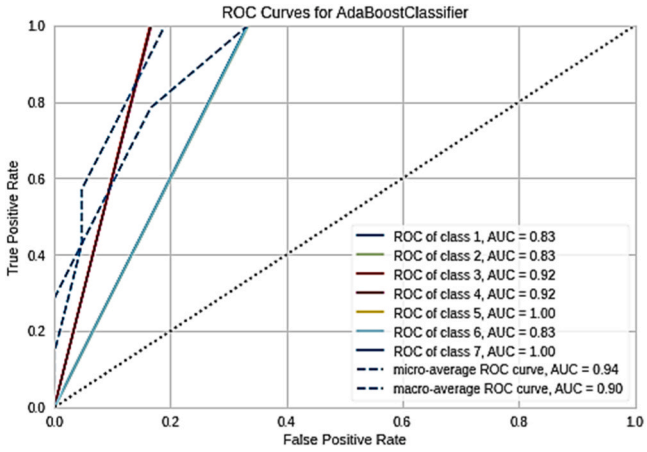


Fig. 8. Learning curve - AdaBoost at Stage-2.

Table 4
Model performance at stage-2.

Models (stage 2)	Acc.	Recall	Precision	F1
Bossvs	0.78	0.79	0.89	0.80
SVM	0.89	0.87	0.92	0.89
Adaboost	0.76	0.7	0.68	0.6
NB	0.93	.94	0.94	0.93
Ridge	0.91	0.92	0.93	0.91
TST	0.92	-	-	-
ANN	**0.93**	**0.93**	**0.94**	**0.93**
GRU	**0.94**	**0.94**	**0.95**	**0.94**
CNN-LSTM	**0.89**	**0.90**	**0.91**	**0.89**
CNN	**0.97**	**0.97**	**0.97**	**0.97**

minimize the error.

CNN - is another deep-learning NN primarily designed to adaptively and automatically learn spatial hierarchy patterns (Yamashita et al., 2018). CNN has a critical aspect of reducing the number of parameters in conventional ANN networks that support the development of larger complex models to solve complicated tasks (Albawi et al., 2017).

GRU - is a simplified and improved LSTM model. It has no cell state, combines input, and forgets gates into a single gate with an update gate, meaning it has fewer gates than LSTM. It also has fewer parameters, which provides a great advantage in terms of convergence and performance (Chung et al., 2014).

LSTM - is a recurrent neural network wherein the long-term dependencies that connect the current task with the previous task do not vanish and are captured by finding appropriate weights. Authors in Althubiti et al. (2018) have evaluated the performance of LSTM against classical models. The accuracy of the LSTM model was achieved 99.97 % compared to 88.83 % with Naïve Bayes algorithm.

RNNs that rely on sequential data are great tools for extracting temporal features. On the other hand, to overcome the gradient disappearing problem, LSTM and GRU are designed. The hybrid model provides a unique advantage, such that CNN can extract spatial and

Table 5
Model performance (different classes' predictions).

STAGE - 2	Class Label	ANN	GRU	CNN-LSTM	CNN
Normal	**1**	✓	✓	✓	✓
Load Change (Incremental)	**2**	×	×	×	×
Line fault (phase 'A')	**3**	×	✓	×	✓
Load Change (Decremental)	**4**	×	×	✓	✓
Sensor Fault	**5**	✓	✓	✓	✓
Cyber-attack (Inverter−1)	**6**	×	✓	✓	✓
Cyber-attack (Inverter−2)	**7**	×	×	×	✓

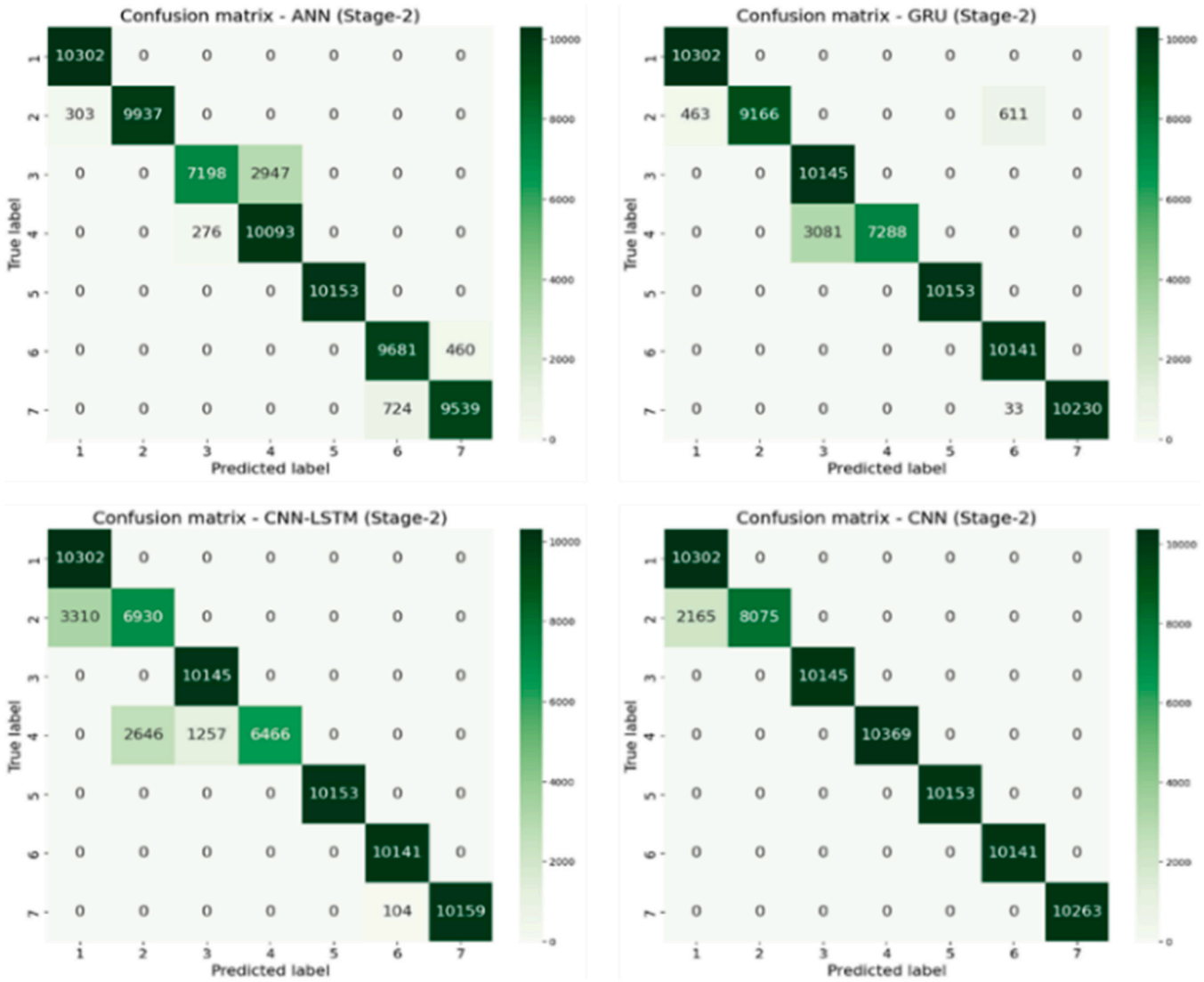


Fig. 9. Confusion matrix at Stage-2.

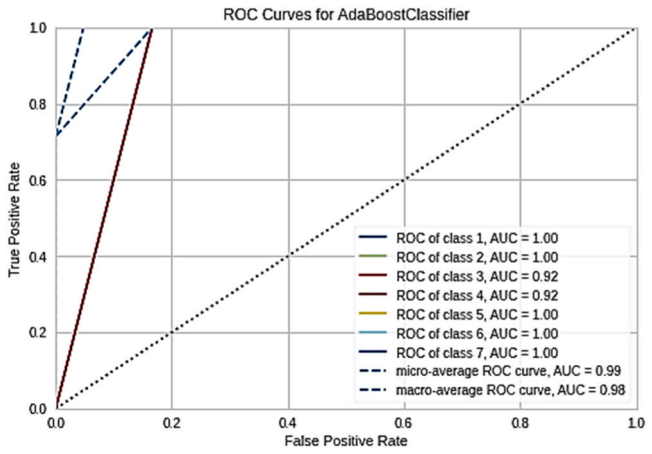


Fig. 10. ROC curve - AdaBoost at Stage-3.

Table 6
Model performance at stage-3.

Models (stage 3)	Acc.	Recall	Precision	F1
Bossvs	0.82	0.81	0.91	0.83
SVM	0.91	0.89	0.92	0.92
Adaboost	0.86	0.86	0.78	0.81
NB	0.94	0.94	0.95	0.94
Ridge	0.90	0.90	0.93	0.90
TST	0.93	-	-	-
ANN	**0.97**	**0.97**	**0.97**	**0.97**
GRU	**0.99**	**0.99**	**0.99**	**0.99**
CNN-LSTM	**0.99**	**0.99**	**0.99**	**0.99**
CNN	**0.99**	**0.99**	**0.99**	**0.99**

global components, and LSTM can extract temporal features.

The hybrid **CNN-LSTM** model benefits from both architectures by first using CNN layers to capture spatial features in network traffic and then processing the sequential patterns in these features with LSTM layers. This combination allows the model to detect multi-stage and evolving attacks that may unfold over time, a capability often lacking in standalone models. This enhanced detection capability makes the hybrid approach particularly suitable for microgrid environments, where detecting nuanced attack vectors in real time is critical.

5. Models interpretation and evaluation

The XAI approach described here interprets model performance to demonstrate how different models learn, train, and predict the probability of classifying events.

Learning curve - is a diagnostic tool used in ML algorithms to illustrate the learning ability of the respective machine learning model. After each update in training, the models are evaluated on the given training dataset and a hold-out validation set. The learning curve represents the measured performance of models in training and cross-validation (CV) tasks.

Prediction probability - is a useful tool for multi-class classification problems where the performance of prediction models in each class is evaluated. A higher probability of predicting a particular class indicates that the model is highly efficient in predicting that class.

Feature importance - is a common method for analyzing the behavior of different ML models based on their preference for features during learning. After evaluating the importance of the features, redundant features are removed, and only the most relevant features are considered for further experiments.

The classification models developed in this study are evaluated on many parameters derived from the confusion matrix, which are as follows (Chen et al., 2021):

Accuracy: $(TP + TN)/(TP + TN + FP + FN)$
Precision: $TP/(TP + FP)$
Recall: $TP/(TP + FN)$
F1 Score: $(2 \times Precision \times Recall)/(Precision + Recall)$
where,
True Positive (*TP*): Predicted correctly the normal situation.
True Negative (*TN*): Predicted correctly the attack classes.
False Positive (*FP*): Predicted incorrectly the normal situation.
False Negative (*FN*): Predicted incorrectly the attack classes.

6. Simulation study, results analysis and discussion

6.1. Quantitative analysis

Stage-1

In Fig. 5(a), it is evident that features V1, V2, and V3 are not important for any of the models in this research. Therefore, in order to improve speed and performance, these features should be removed.

In Fig. 5(b), the training and cross-validation accuracies are very close to each other, indicating that there isn't much model variance. The accuracy increases as the number of training samples increases. However, we can't determine the impact of individual class sample sizes on model accuracy from this curve.

Fig. 6 shows the ROC curve for the AdaBoost classifier, which demonstrates the model's performance on different classes and its improvement over successive stages. The class imbalance significantly affects the curve, with classes having fewer samples exhibiting lower AUC scores and tending to be closer to the diagonal line.

In Table 2, the performance of various models is summarized across different metrics. CNN and NB exhibit higher accuracy levels of 97 % and 92 %, respectively. The classical model CNN demonstrates a high precision rate of 98 % along with a 97 % F1 score. As highlighted in Table 2, the ANN and AdaBoost models in Stage-1 show the lowest accuracy and require the longest training time (refer to Fig. 12). In contrast, the CNN model achieves the highest accuracy and requires the least training time. It's worth noting that the hybrid model CNN-LSTM does not perform well in Stage-1 when dealing with unbalanced data.

According to the confusion matrix (Fig. 7), all Stage-1 models effectively predict normal conditions (no event-class 1). The ANN and GRU models demonstrate strong predictions for line fault events (class-3), while the CNN-LSTM model excels in predicting decremental load change events (class-4) and cyber-attacks in inverter 2 (class-7). However, the CNN model outperforms the others in Stage 1, accurately predicting line fault events (class-3), sensor faults in a microgrid (class-5), decremental load changes (class-4), and cyber-attacks in inverter 2 (class-7), as summarized in Table 3.

Stage-2

In this experiment, Fig. 8 depicts the ROC curves for the Ada Boost classifier in stage 2. The curve illustrates that after data augmentation and feature engineering, there has been an improvement in the accuracy of the respective classes. The overall performance AUC score for AdaBoost is 90 % in stage 2, compared to 89 % in stage 1.

As summarised in Table 4, the CNN-LSTM model in Stage-2 has the least accuracy, though it improved by over 5.65 % compared to Stage-1. ANN model significantly improved by around 26 % accuracy compared to Stage-1. However, the CNN model has a slight decrement in accuracy of 0.9 % with respect to Stage-1 but still performs best among all models. The class imbalance solution through the SMOTE-ENN algorithm seems to improve the classification accuracy. The computational time for training the models is highest at 16 min for GRU in Stage-2, and CNN is very fast and is trained in 7 minutes among all other models, as shown in (Fig. 12).

As shown in the confusion matrix in Fig. 9, all models in Stage-2 are perfect in predicting the normal condition (no event-class1). The ANN, which could predict line fault correctly in Stage-1, now has a false

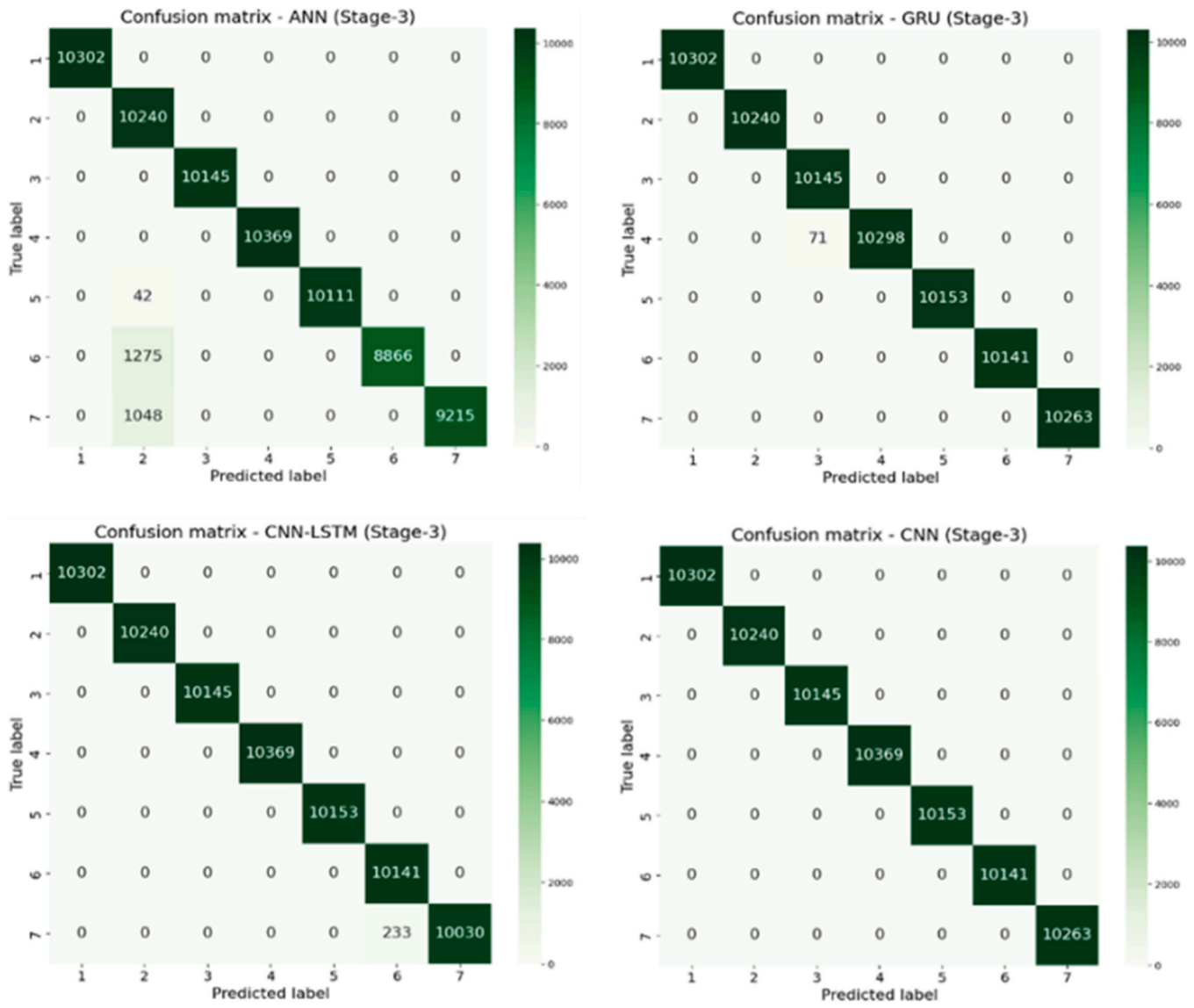


Fig. 11. Confusion matrix at Stage-3.

Table 7
Model performance (different classes' predictions).

STAGE - 3	Class Label	ANN	GRU	CNN-LSTM	CNN
Normal	**1**	✓	✓	✓	✓
Load Change (Incremental)	**2**	✓	✓	✓	✓
Line fault (phase 'A')	**3**	✓	✓	✓	✓
Load Change (Decremental)	**4**	✓	×	✓	✓
Sensor Fault	**5**	×	✓	✓	✓
Cyber-attack (Inverter−1)	**6**	×	✓	✓	✓
Cyber-attack (Inverter−2)	**7**	×	✓	×	✓

prediction for line fault but can correctly predict microgrid sensor fault. GRU model can correctly predict line fault, same as stage-1, but in stage-2, it can also predict microgrid sensor fault and cyber-attack in inverter 1. CNN-LSTM model, now in Stage-2, has false predictions in decremented load change and cyber-attack in inverter 2 but has all labels predicted correctly for microgrid sensor fault and cyber-attack in inverter 1. CNN model in Stage-2 has all labels correctly classified for all events except the event of incremental load change, as summarised in Table 5.

Stage-3

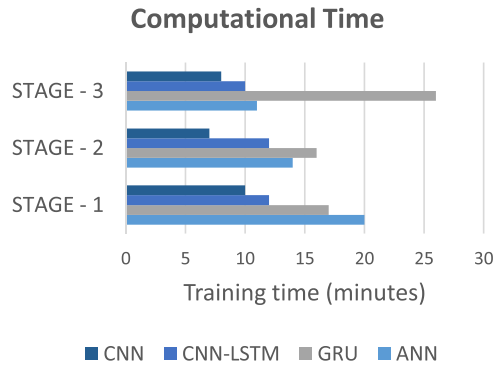


Fig. 12. Required computational time.

The ROC curve for the Stage-3 experiment, as depicted in Fig. 10, indicates that only classes 3 and 4 have an AUC score of around 92 %, while all the other classes are perfectly classified, resulting in an overall AUC score of 97.62 %, compared to 89.6 % in Stage-1. The ROC curves presented in the paper for the AdaBoost classifier in each stage suggest that the classification accuracy for classes with lower samples improves

Table 8
comparison of Model performance with other literature.

Model	Reference	Accuracy	This Paper
SVM	(Tuan et al., 2020)	0.92	0.91
ANN	(Ahmad et al., 2021; Subba et al., 2016)	0.95	0.97
GRU	(Kaur et al., 2023; Rehman et al., 2021)	0.95	0.99
CNN-LSTM	(Khan et al., 2021; Vinayakumar et al., 2017)	0.98	0.99
CNN	(Kaur et al., 2023)	0.85	0.99

in the successive stages after applying data augmentation, feature extraction, and hyper-parameter tuning.

According to Table 6, all models have shown an improvement in accuracy compared to previous stages. NB and ANN have an accuracy of 94 % and 96.68 %, respectively, while GRU and CNN-LSTM have an accuracy of 99.9 % and 99.67 %, respectively. After hyper-parameter tuning in Stage-3, CNN achieves the highest accuracy of 99.99 % with all events correctly classified, even though it had less accuracy in Stage-2 compared to Stage-1.

Fig. 11 (confusion matrix) shows that all models in Stage-3 accurately predict normal conditions, with no event-class 1. The ANN model correctly identifies incremental load change, line fault in phase 'A', and decremented load change but has many false predictions for other classes. The GRU model accurately classifies all events except decremented load change. The CNN-LSTM model has inaccurate predictions only for cyber-attacks in Inverter 2 events, but correctly predicts all other events. The CNN model accurately predicts all labels and is the best among all models, which is also summarised in Table 7.

The computational time for training the models is highest for GRU in Stage-3, which takes 26 minutes, while CNN is the fastest and only takes 8 minutes among all other models. Additionally, the tuned models have less computational burden compared to previous stages. Only GRU takes longer in Stage-3, whereas other models have a reduction in the training time, as shown in Fig. 12.

When comparing the results of the other literature with the findings presented in this paper, Table 8 shows notable differences in accuracy across various models. For the Support Vector Machine (SVM) model, accuracy in reference stands at 0.92, while this paper reports a slightly lower accuracy of 0.91. Moving to the Artificial Neural Network (ANN), references (Ahmad et al., 2021; Subba et al., 2016) cite an accuracy of 0.95. However, the results obtained in this paper surpass expectations

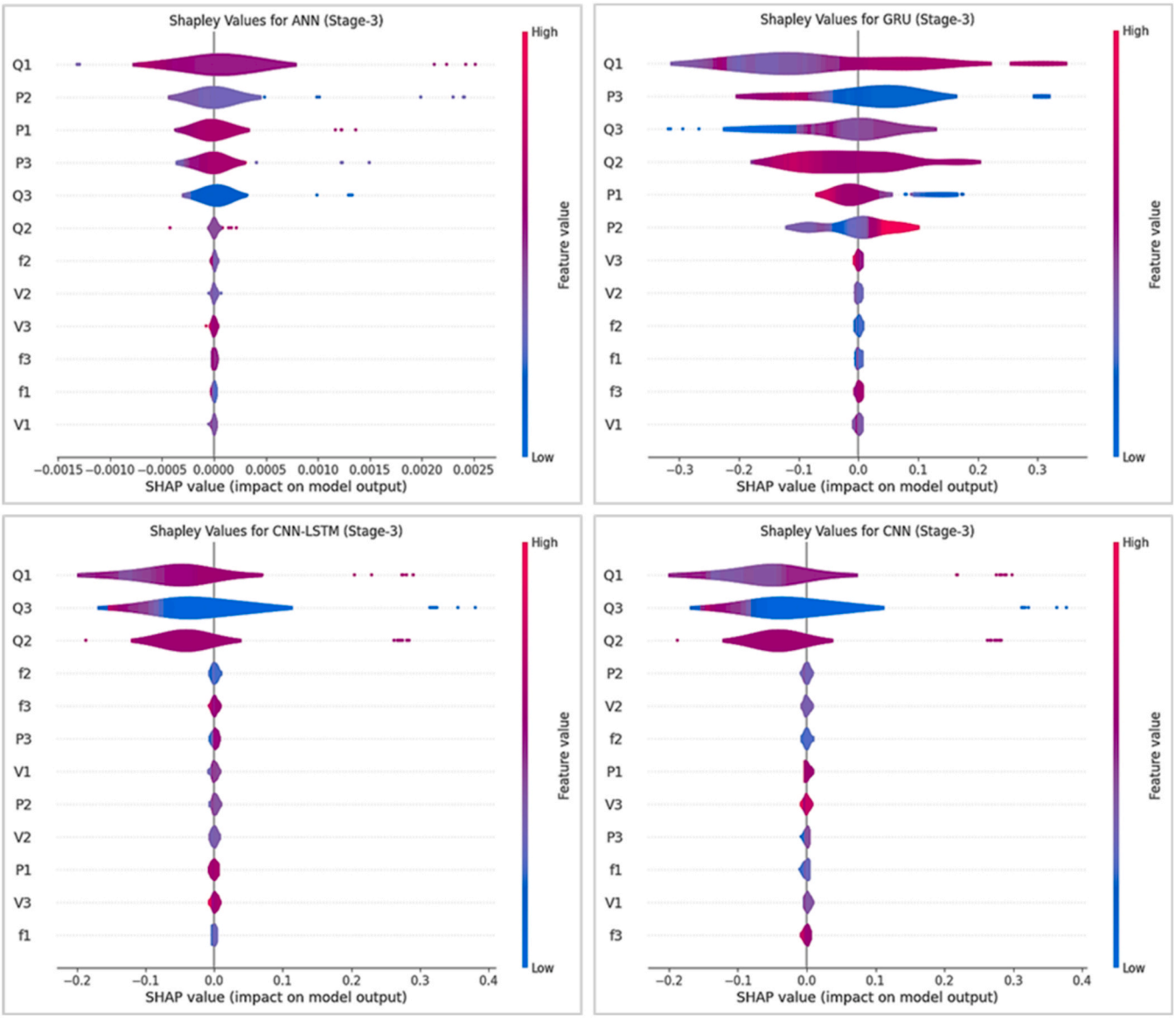


Fig. 13. SHAP plot for feature importance at Stage-3.

with an accuracy of 0.97, indicating an improvement over the referenced performance.

For the Gated Recurrent Unit (GRU) model, the accuracy in Kaur et al. (2023), Rehman et al. (2021) is documented as 0.95. Remarkably, the findings in this paper showcase a significant enhancement, achieving an accuracy of 0.99. In the case of the Convolutional Neural Network (CNN)-Long Short-Term Memory (LSTM) model, the accuracy in Khan et al. (2021), Vinayakumar et al. (2017) reported as 0.98. However, this paper demonstrates a matching accuracy of 0.99, indicating consistency with the referenced performance. Lastly, for the CNN model, the reference (Kaur et al., 2023) accuracy is notably lower at 0.85. In contrast, this paper presents a substantially improved accuracy of 0.99, suggesting a considerable enhancement in model performance compared to the referenced study.

6.2. Qualitative analysis

6.2.1. Model interpretation: SHAP technique (XAI)

The Shapley value explanation method possesses several attractive properties, such as local accuracy, missingness, and consistency (Lundberg and Lee, 2017). It decomposes the prediction into a linear combination of feature contributions (Lundberg et al., 2018), as illustrated below.

$$g(z') = \quad \varnothing_0 + \sum_{j=1}^{p} \varnothing_j z'_j$$

where g is the explanation method, p is the number of features, and $z' \in \{0,1\}^p$ is the coalition vector that indicates the on or off state of each feature. The Shapley value of the j-th feature is denoted as $\varnothing_j$, which can be estimated using shap Python package (PiML). The exact solution is obtained using the Shapley value formula, which requires evaluating all possible coalitions of features with and without the i-th feature.

$$\varnothing_i = \sum_{S \quad \{1,\ldots,p\}\{i\}} \frac{|S|!(p-|S|-1)!}{p!}(val(S \cup \{i\}) - val(S)))$$

where val is the value function that returns the prediction of each coalition. The marginal contribution of feature i to the coalition S is calculated as the difference between the value of the coalition with the addition of feature i and the value of the original coalition, i.e., $(val(S \cup \{i\}) - val(S)$. The term $\frac{|S|!(p-|S|-1)!}{p!}$ is a normalization factor. When the number of features is small, this exact estimation approach is acceptable. However, as the number of features increases, the exact solution may become problematic.

It's worth noting that the value function *val* takes the feature coalition S as input. However, in machine learning models, the prediction is not solely based on the feature coalition but on the entire feature vector. KernelSHAP is a kernel-based approach for estimating Shapley values that are inspired by local surrogate models. It is particularly useful for computing Shapley values with a large number of features. Given an instance x, KernelSHAP estimates the contributions of each feature using the following steps:

- Randomly sample coalitions and compute the output of these simulated coalitions.
- Calculate the weight of each feature in the coalition using the SHAP Kernel).
- Fit a weighted linear model to the sampled coalitions.
- Return the Shapley values which are the coefficients of the linear model.

In a Shapley summary violin plot, each feature is represented by a violin-shaped curve that shows the density of its Shapley values. The width of the curve indicates the density of observations with different values of the feature, while the height indicates the density of Shapley values for those observations.

In this research, SHAP values for all models in Stage-3 are considered to analyse the kernel-based impact of features on model output. A feature importance plot using SHAP values is shown in Fig. 13. For ANN, reactive and active power (Q1, P2, P1, P3, Q3, and Q2- in descending order) have a huge impact on the model output with some outliers for Q1 on both positive and negative SHAP values. It also appears that most of the SHAP values for all features in the ANN model lie around zero. For the GRU model, Q1 has high positive SHAP values affecting the model predictability; however, most SHAP values for Q1 lie below zero and have a very low impact. P3 and P2 have negative and positive SHAP values, respectively, affecting the model output. In the CNN-LSTM hybrid model, only reactive power values (Q1, Q3, Q2 in descending order) are highly impactful, with most of the SHAP values being negative, but a few outliers in Q1 and Q2 have high-impact positive values. The CNN models also have similarities with a hybrid model, with the same feature importance, but the order of feature impact changes after the top 3 features. In hybrid model, the descending order gives preference to f2, f3, P3, V1, P2, V2, P1, V3, and f1 but in CNN model, the order follows as P2, V2, f2, P1, V3, P3, f1, V1 and f3.

7. Conclusion

This framework for detecting cyber-attacks is data-centric and systematically focuses on improving model performance. Stage-1 involves collecting and pre-processing data, while Stage-2 improves data quality and extracts features. Stage-3 entails enhancing models through hyper-parameter tuning. Models are analyzed at every stage, and better insights can be gained by interpreting their learning behavior. The first stage's model interpretation revealed that classes with fewer observations had lower accuracy and prediction probability across all models. Additionally, feature importance scores indicate the features that each model should consider. Data augmentation and hyper-parameter tuning also show how the model accuracy is improved. The framework has been implemented in a microgrid network to demonstrate how XAI improves cyber-attack detection capability. Among all considered models, GRU, CNN, and CNN-LSTM performed the best (99 %). The new TST model developed for cyber-attack classification also had an accuracy of 93 %.

CRediT authorship contribution statement

Rohit Trivedi: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Shafi Khadem:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Sandipan Patra:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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